Awakened Rotation Cosmology:
Toward a Self-Consistent Resolution of the CMB–BAO Tension

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Abstract

We present a structural reformulation of cosmological dynamics motivated by persistent inconsistencies between early- and late-time observables. Building on a rotation-based modification of photon microphysics that successfully reproduces key features of the cosmic microwave background (CMB), we identify a fundamental limitation in background expansion models of the form $H(z) \propto (1 + z)$.

We argue that resolving the mismatch between CMB-derived and baryon acoustic oscillation (BAO) scales requires abandoning direct parametrizations of $H(z)$ in favor of a dynamical variable framework. We introduce a recursive dynamical variable $\theta(z)$, whose evolution governs the expansion history through a Friedmann-like equation.

We further propose that cosmological dynamics may be influenced by an external gravitational environment ("gravitational bath"), which acts not as a primary driver but as a boundary condition for the recursive system.

This work establishes the mathematical structure of the problem and outlines a pathway toward a fully self-consistent cosmological model.

1 Introduction

The standard cosmological model successfully describes a wide range of observations, yet tensions remain between early-universe probes (e.g., CMB) and late-time measurements (e.g., BAO and supernova distances).

In particular, the consistency of the sound horizon scale across epochs imposes strong constraints on the expansion history $H(z)$. Any deviation in early-time dynamics propagates into measurable discrepancies at low redshift.

This work originates from an attempt to modify photon-sector microphysics through a redshift-dependent rotational factor $F_{\mathrm{rot}}(z)$, which successfully reproduces:

- Silk damping behavior
- Polarization amplitudes (EE, TE)
- Integrated Sachs–Wolfe (ISW) trends

However, this success at the level of perturbations does not extend to background observables. In particular, models with $H(z) = H_0(1 + z)$ exhibit structural incompatibility with BAO and supernova data.

This suggests that the problem is not one of parameter tuning, but of underlying dynamical structure.

2 Breakdown of Direct Expansion Models

We demonstrate that any model based on a fixed functional form of $H(z)$, even with additional components, encounters a structural tradeoff:

- Matching the sound horizon r_s requires large modifications to high-redshift expansion
- Matching luminosity distances requires low-redshift behavior close to ΛCDM

These constraints cannot be simultaneously satisfied within additive or multiplicative modifications of $H(z)$.

This indicates that:

> The expansion rate must emerge from a deeper dynamical system rather than being directly prescribed.

3 Recursive Dynamical Variable

We introduce a dynamical variable $\theta(z)$, interpreted as a cumulative phase associated with cosmological evolution.

The expansion rate is not specified directly, but arises from a Friedmann-like relation:

$$H^2(z) = \left(\frac{d\theta}{dz}\right)^2 + V(\theta) \tag{1}$$

where:

- $\theta(z)$ is a state variable
- $V(\theta)$ is an effective potential

This structure is analogous to scalar-field cosmology, but with a key distinction: θ is not introduced as a matter field, but as a geometric or kinematic degree of freedom.

4 Emergent Matter and Vacuum Components

Empirically, we find that a potential of the form:

$$V(\theta) \approx H_0^2 \left[\Omega_m e^{3\alpha\theta} + \Omega_\Lambda\right] \tag{2}$$

reproduces key features of standard cosmology.

Notably:

- An effective matter density $\Omega_m \approx 0.3$ emerges from the dynamics
- A constant term behaves like vacuum energy

This suggests that matter and dark energy may arise as effective components of a deeper rotational-recursive system.

5 Separation of Micro and Macro Sectors

The framework naturally separates into two layers:

Micro (CMB sector)

Controlled by $F_{\mathrm{rot}}(z)$, affecting scattering, diffusion, and anisotropies.

Macro (geometry sector)

Controlled by $\theta(z)$ dynamics, determining $H(z)$, distances, and BAO observables.

This separation resolves previous conflicts where a single function was required to satisfy incompatible constraints.

6 Gravitational Bath as Boundary Condition

We propose that the cosmological system may not be fully closed, but embedded in a larger gravitational environment.

In this interpretation:

- Recursive dynamics define internal evolution
- The external "gravitational bath" sets boundary conditions

These may include normalization of H_0 and initial conditions for θ.

7 Open Problem (B-task)

The central unresolved problem is:

> Determine the fundamental Lagrangian governing $\theta(z)$ such that the resulting $H(z)$ simultaneously satisfies CMB, BAO, and supernova constraints.

This requires:

- Self-consistent solution of coupled equations for $\theta(z)$ and $H(z)$
- Identification of the physical origin of $V(\theta)$
- Clarification of the role of boundary conditions

8 Discussion

The key outcome of this work is a restructuring of the problem:

- Direct parametrizations of $H(z)$ are insufficient
- A dynamical-variable approach is required
- Microphysical and geometric sectors must be separated
- External embedding may play a secondary role

9 Conclusion

We have shown that:

- Rotation-based modifications successfully address CMB microphysics
- Background expansion requires a dynamical formulation
- A recursive variable $\theta(z)$ provides a viable framework
- Matter and vacuum energy may emerge from underlying dynamics

The full solution requires determining the governing Lagrangian, which remains an open problem.